



Lab 01: GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Name: Sidra Iftikhar

Reg no: FA24B1-SE-063

Course code: SE-665

Section: A

GitHub Username: engineer-sidra-iftikhar

Task 1: Create and Configure Your GitHub Account

GitHub Profile:

Engr Sidra Iftikhar

engineer-sidra-iftikhar · she/her

🎓 SE Student @ FJWU · 4th Semester ·
CGPA 3.40 📚 Databases · Networking ·
UML & System Design 🔧 C++ · SQL ·
Streamlit · Cisco Packet Tracer 🌐 GDG
Cloud Isl

Edit profile

🏢 FJWU
📍 Islamabad, Pakistan
✉️ engineersidra58@gmail.com
🔗 <https://linkedin.com/in/engr-sidra-rajput>
🌐 in/engr-sidra-rajput

Screenshot: GitHub Profile

GitHub actual name:

Public profile

Name

Engr Sidra Iftikhar

Your name may appear around GitHub where you contribute or are mentioned. You can remove it at any time.

Screenshot: GitHub actual name

Task 2: Register in the Student Marks Portal

Portal login page:

ID

Complete registration

Enter your registration number, course and section.
The enrollment requires approval.

Registration number

2024-BSE-63 ▼

Allowed: 2024-BSE-01 to 2024-BSE-99.

Course

SE-665 — Cloud Computing ▼

Section

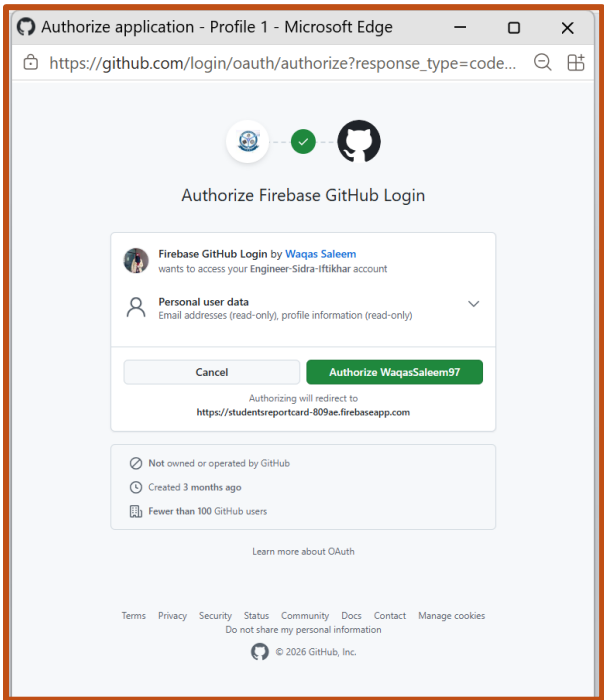
Section A ▼

Submit enrollment

Sign out

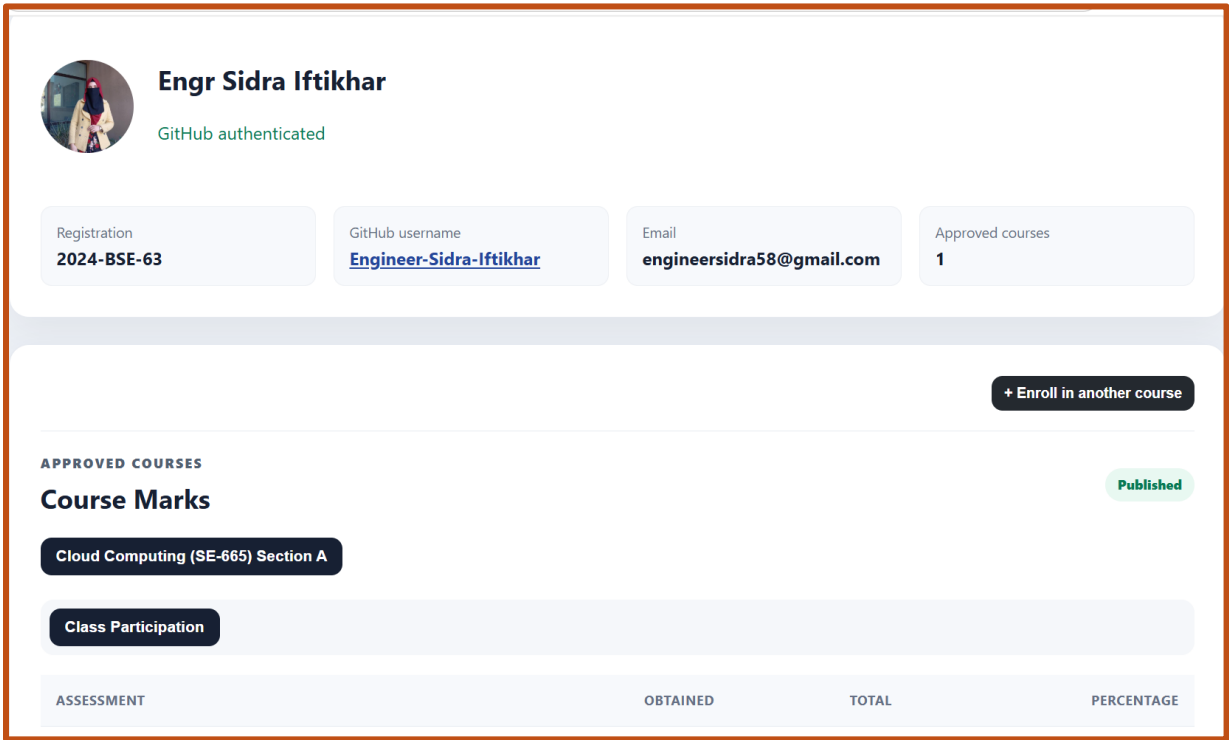
Screenshot: Portal login page

Portal GitHub profile:



Screenshot: Portal GitHub profile

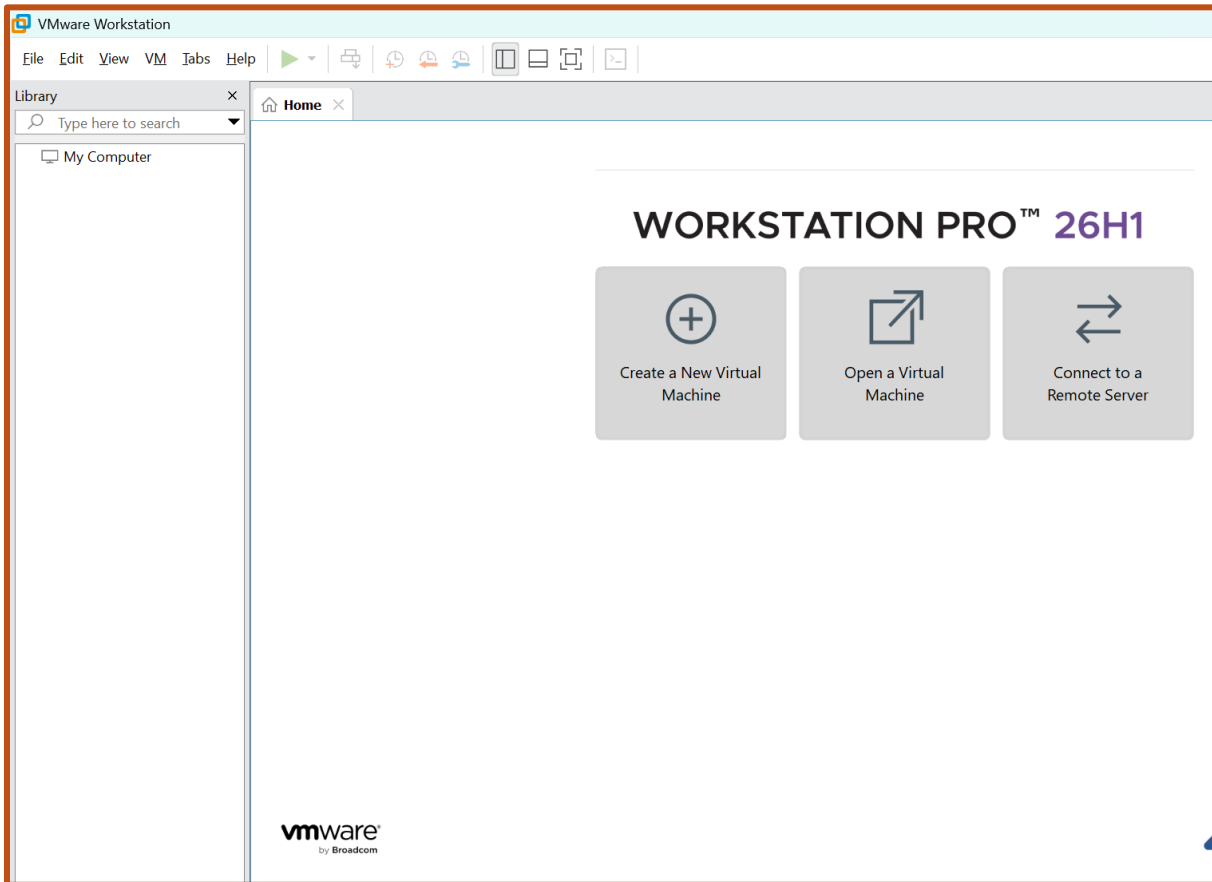
Portal enrollment submitted:



Screenshot: Portal enrollment submitted

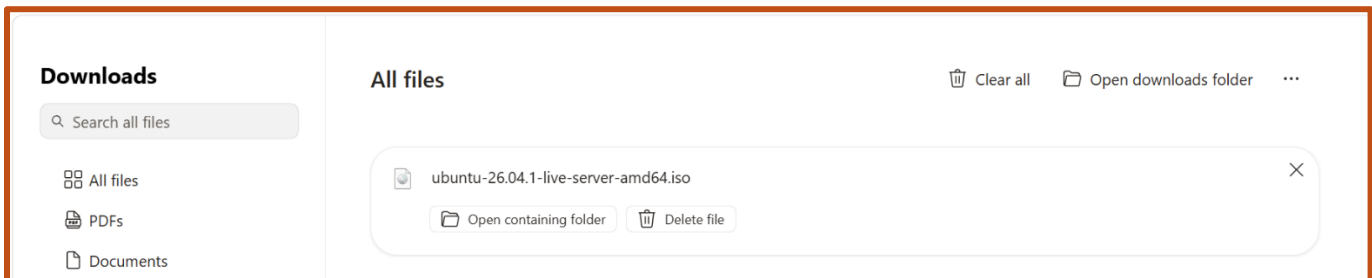
Task 3: Download VMware Workstation Pro and Ubuntu Server

VMware WorkStation:



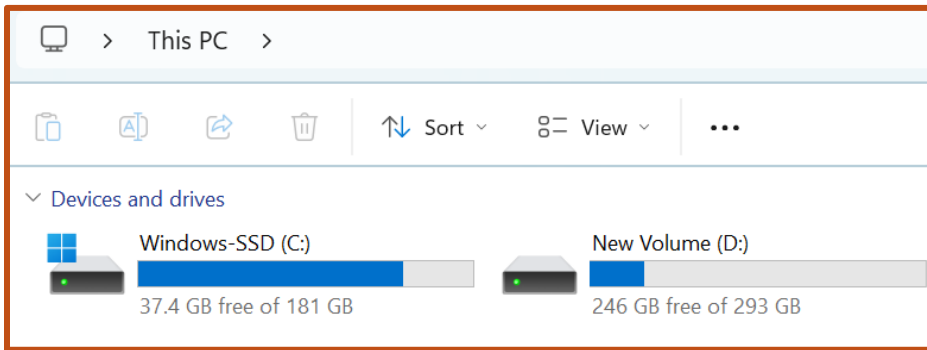
Screenshot: VMware WorkStation

ubuntu_server_iso:



Screenshot: Ubuntu server iso

available_storage:



Screenshot: Available storage

Task 4: Create the Ubuntu Server Virtual Machine

vm_creation_wizard:



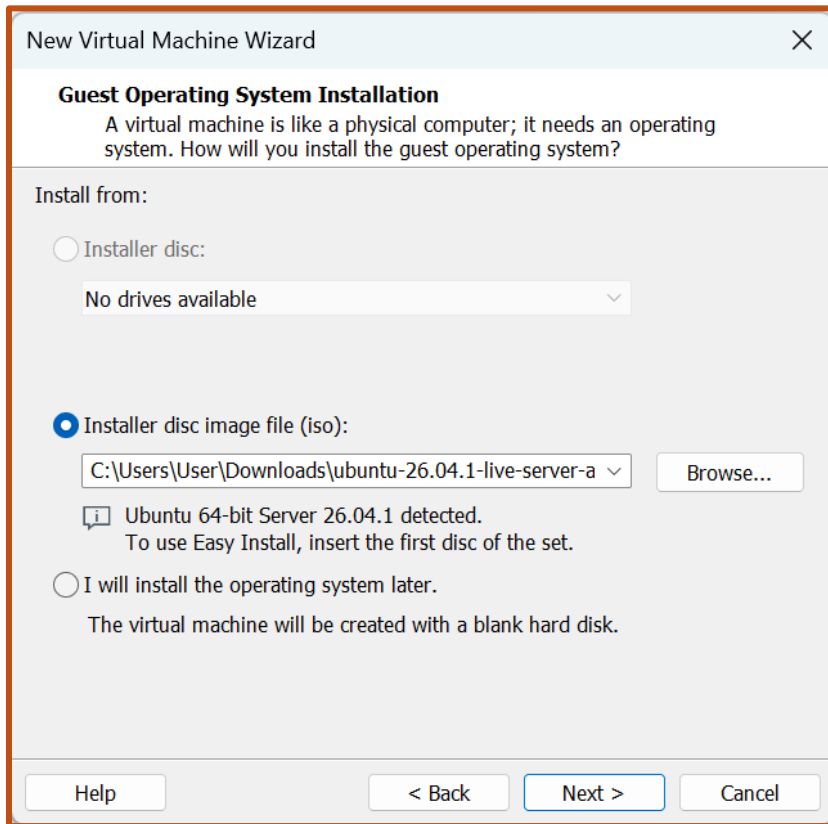
Screenshot: Vm creation wizard

vm_typical_configuration:



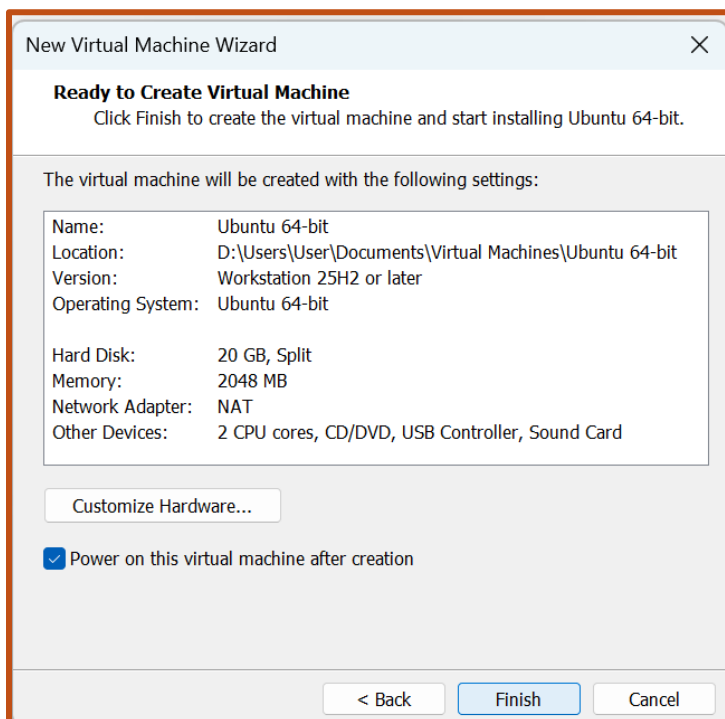
Screenshot: Vm typical configuration

vm_ubuntu_iso_selected:



Screenshot: Vm ubuntu iso selected

vm_configuration_summary:



Screenshot: Vm configuration summary

ubuntu_installer_boot:

```
Starting snapd.apparmor.service - ...files managed internally by snapd...
Starting systemd-binfmt.service - Set Up Additional Binary Formats...
Starting systemd-tmpfiles-setup.service - Create System Files and Directories...
Starting ufw.service - Uncomplicated firewall...
[ OK ] Finished finalind.service - Create runtime dir for shutdown pivot root.
[ OK ] Finished plymouth-read-write.service - Plymouth To Write Out Runtime Data.
[ OK ] Finished ufw.service - Uncomplicated firewall.
[ OK ] Finished systemd-udev-trigger.service - Coldplug All udev Devices.
[ OK ] Started systemd-ask-password-console.asks to Console Directory Watch.
[ OK ] Reached target cryptsetup.target - Local Encrypted Volumes.
[ OK ] Finished kdump-tools.service - Kernel crash dump capture service.
[ OK ] Finished systemd-tmpfiles-setup.service - Create System Files and Directories.
Starting ldconfig.service - Rebuild Dynamic Linker Cache...
Starting systemd-journal-catalog-update.service - Rebuild Journal Catalog...
[ OK ] Started vgauth.service - Authentic... virtual machines hosted on VMware.
[ OK ] Started open-vm-tools.service - Se... virtual machines hosted on VMware.
Mounting proc-sys-fs-binfmt_misc.mount - xecutable File Formats File System...
[ OK ] Mounted proc-sys-fs-binfmt_misc.mount - xecutable File Formats File System.
[ OK ] Finished systemd-binfmt.service - Set Up Additional Binary Formats.
[ OK ] Finished systemd-journal-catalog-update.service - Rebuild Journal Catalog.
[ OK ] Finished netplan-config.service - Netplan Backend Configuration.
[ OK ] Finished snapd.apparmor.service - ...files managed internally by snapd.
[ 22.151894] piix4_smbus 0000:00:07:3: SMBus Host Controller not enabled!
[ OK ] Finished console-setup.service - Set console font and keymap.
[ OK ] Listening on systemd-rfkill.socket - Load/Save RF Kill Switch Status /dev/rfkill Watch.
[ OK ] Reached target sound.target - Sound Card.
[ OK ] Started cloud-init-main.service - Cloud-Init: Single Process.
Starting cloud-init-local.service - Cloud-Init: Local Stage (pre-network)...
[ OK ] Finished ldconfig.service - Rebuild Dynamic Linker Cache.
Starting systemd-update-done.service - Update is Completed...
[ OK ] Finished systemd-update-done.service - Update is Completed.
[ OK ] Finished cloud-init-local.service - Cloud-Init: Local Stage (pre-network).
[ OK ] Reached target network-pre.target - Preparation for Network.
[ OK ] Listening on systemd-networkd-resolve-hook.socket - Network Management Resolve Hook Socket.
Starting cloud-init-network.service - Cloud-Init: Network Stage...
Starting systemd-networkd.service - Network Management...
[ OK ] Started systemd-networkd.service - Network Management.
[ OK ] Reached target network.target - Network.
Starting systemd-networkd-persistent-storage.service - Enable Persistent Storage in systemd-networkd...
Starting systemd-networkd-wait-online.service - Wait for Network to be Online...
[ OK ] Finished systemd-networkd-persistent-storage.service - Enable Persistent Storage in systemd-networkd.
[ OK ] Finished systemd-networkd-wait-online.service - Wait for Network to be Online.
```

Screenshot: Ubuntu installer boot

Task 5: Choose the Installation Language

ubuntu_language:

Willkommen! Bienvenue! Welcome! Добро пожаловать! Welkom!

[Help]

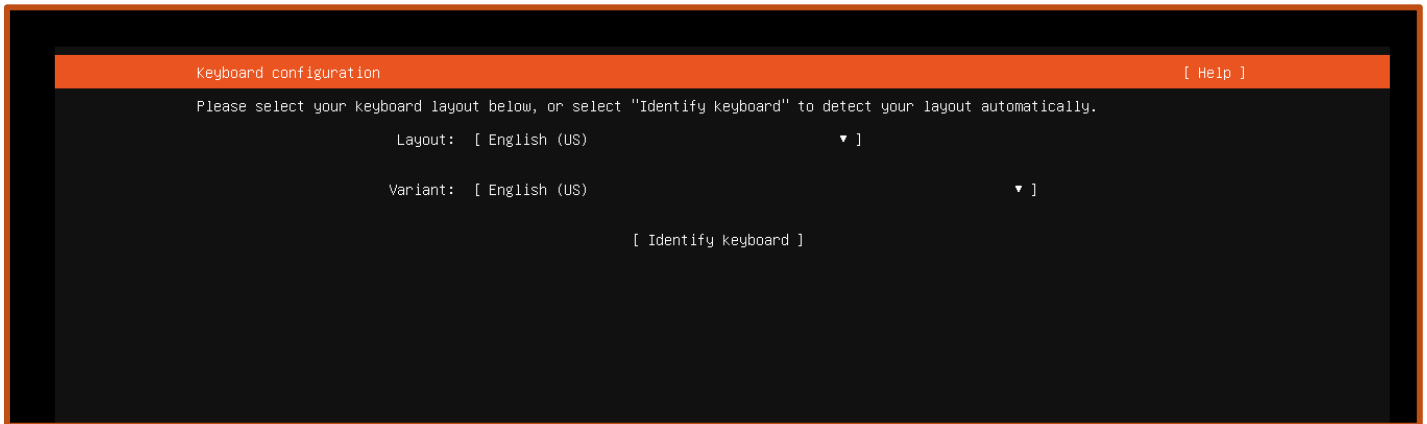
Use UP, DOWN and ENTER keys to select your language.

```
[ Asturianu ]
[ Bahasa Indonesia ]
[ Català ]
[ Deutsch ]
[ English ]
[ English (UK) ]
[ Español ]
[ Français ]
[ Galego ]
[ Hrvatski ]
[ Latviski ]
[ Lietuviškai ]
[ Magyar ]
[ Nederlands ]
[ Norsk bokmål ]
[ Occitan ]
[ Polski ]
[ Português ]
[ Suomi ]
[ Svenska ]
[ Cestina ]
[ Ελληνικά ]
[ Беларуская ]
[ Русский ]
[ Српски ]
[ Українська ]
```

Screenshot: Ubuntu language

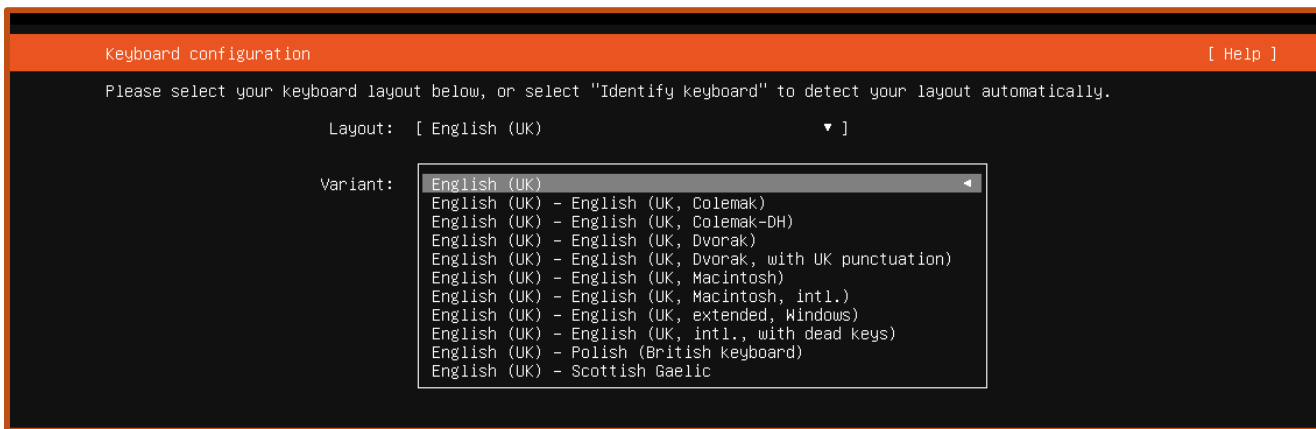
Task 6: Configure the Keyboard Layout

ubuntu_keyboard_layout:



Screenshot: Ubuntu keyboard layout

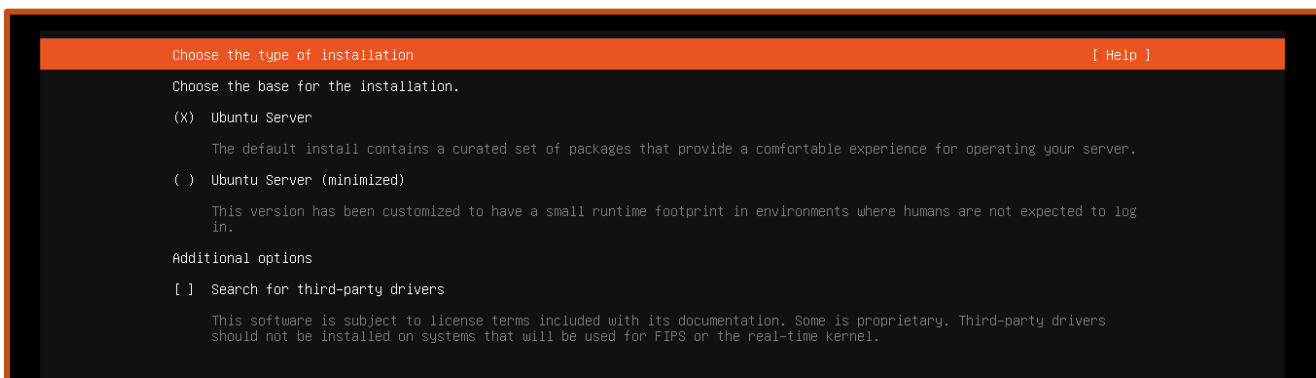
ubuntu_keyboard_variant:



Screenshot: Ubuntu keyboard variant

Task 7: Choose the Ubuntu Server Installation Type

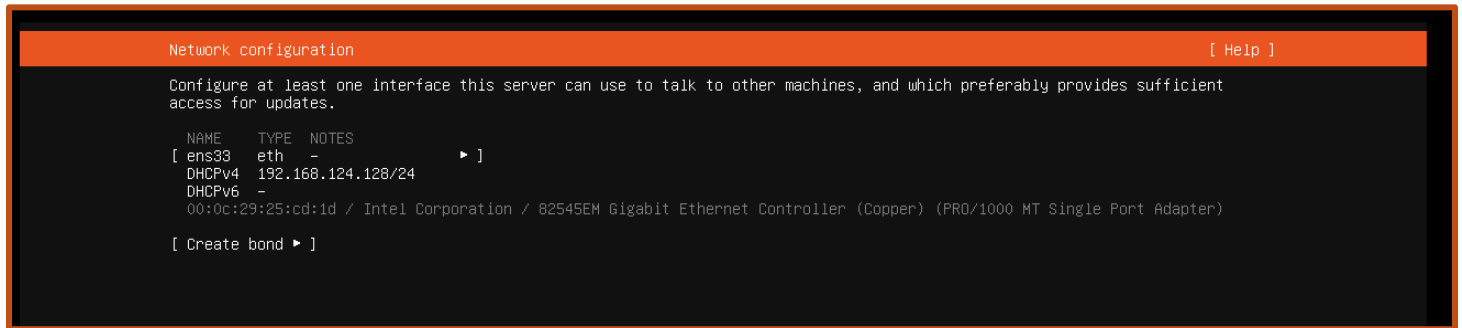
ubuntu_installation_type:



Screenshot: Ubuntu installation type

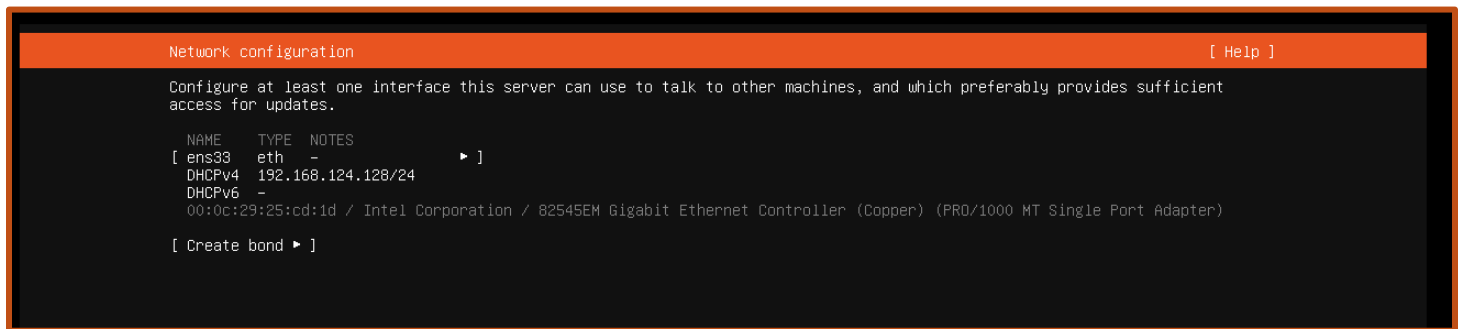
Task 8: Configure the Network

ubuntu_network_interface:



Screenshot: Ubuntu network interface

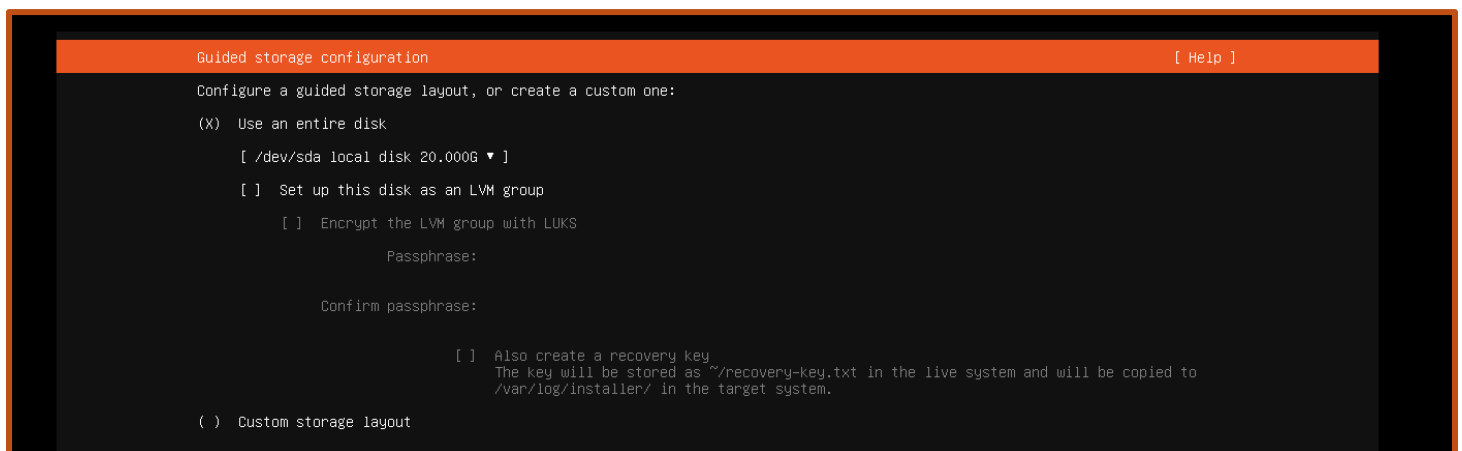
ubuntu_installer_network:



Screenshot: Ubuntu installer network

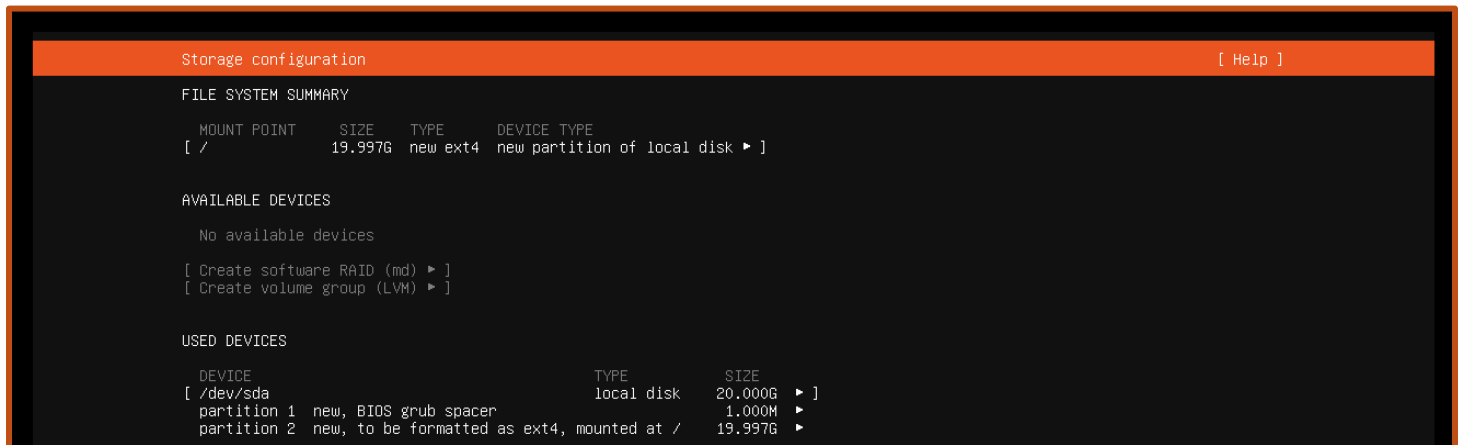
Task 9: Configure Storage

ubuntu_guided_storage:



Screenshot: Ubuntu guided storage

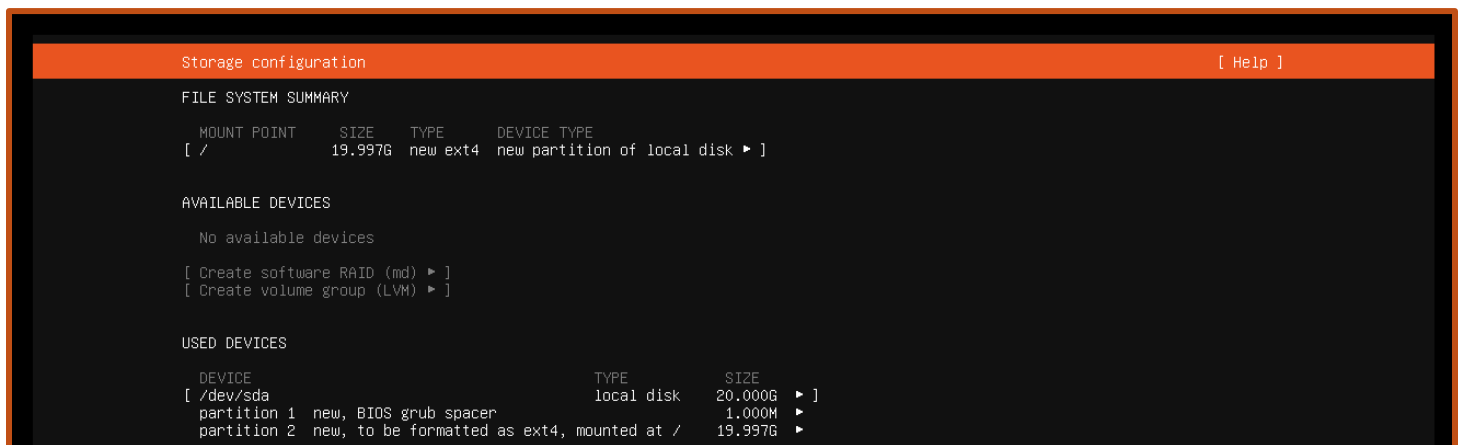
ubuntu_storage_configuration



Screenshot: Ubuntu storage configuration

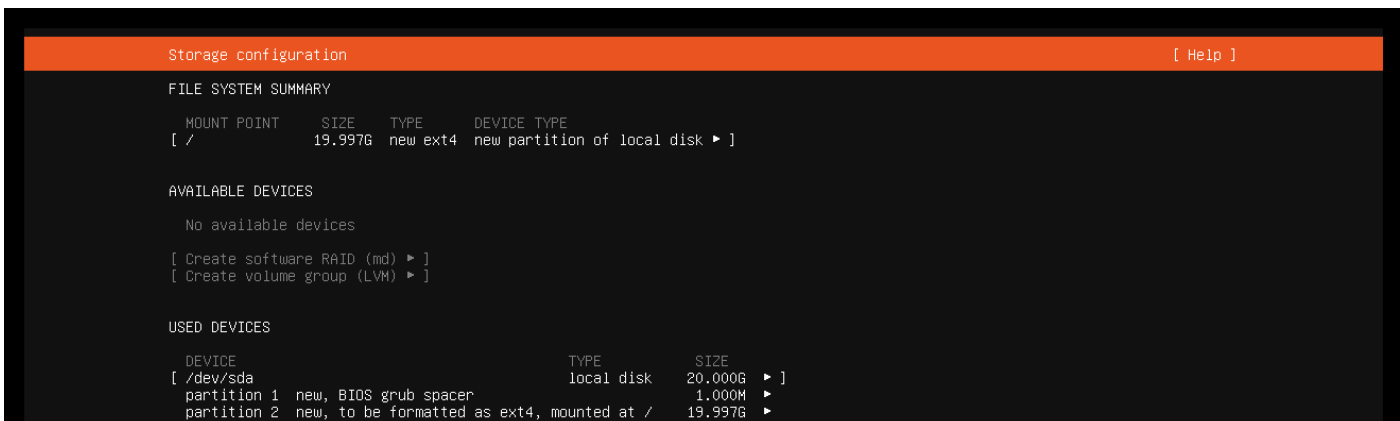
Task 10: Select the Installation Disk

ubuntu_selected_disk



Screenshot: Ubuntu selected disk

ubuntu_disk_details



Screenshot: Ubuntu disk details

Task 11: Review the Partitions

ubuntu_partition_layout:

```
Storage configuration [ Help ]

FILE SYSTEM SUMMARY

MOUNT POINT    SIZE    TYPE    DEVICE TYPE
[ /            19.997G new ext4 new partition of local disk ▶ ]

AVAILABLE DEVICES

No available devices

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]

USED DEVICES

DEVICE                                TYPE    SIZE
[ /dev/sda                             local disk 20.000G ▶ ]
partition 1 new, BIOS grub spacer         1.000M ▶ ]
partition 2 new, to be formatted as ext4, mounted at / 19.997G ▶ ]
```

Screenshot: Ubuntu partition layout

Task 12: Confirm the Storage Changes

ubuntu_storage_warning

```
Storage configuration [ Help ]

FILE SYSTEM SUMMARY

MOUNT POINT    SIZE    TYPE    DEVICE TYPE
[ /            19.997G new ext4 new partition of local disk ▶ ]

AVAILABLE DEVICES

No available devices

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]

USED DEVICES

DEVICE                                TYPE    SIZE
[ /dev/sda                             local disk 20.000G ▶ ]
partition 1 new, BIOS grub spacer         1.000M ▶ ]
partition 2 new, to be formatted as ext4, mounted at / 19.997G ▶ ]

Confirm destructive action

Selecting Continue below will begin the installation process and
result in the loss of data on the disks selected to be formatted.

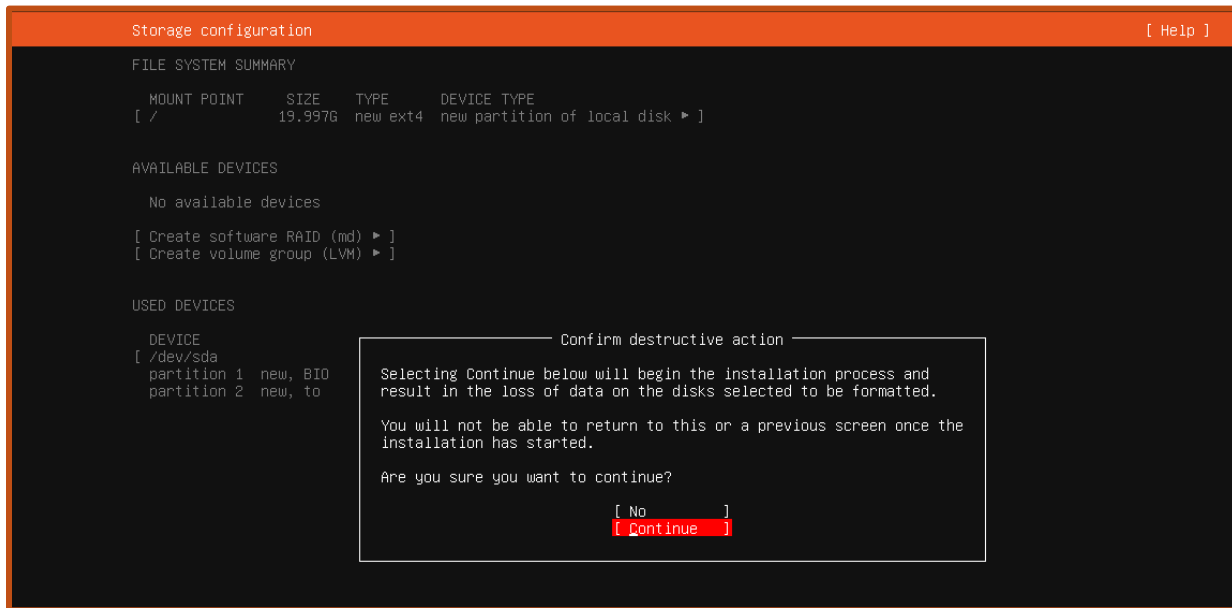
You will not be able to return to this or a previous screen once the
installation has started.

Are you sure you want to continue?

[ No ]
[ Continue ]
```

Screenshot: Ubuntu storage warning

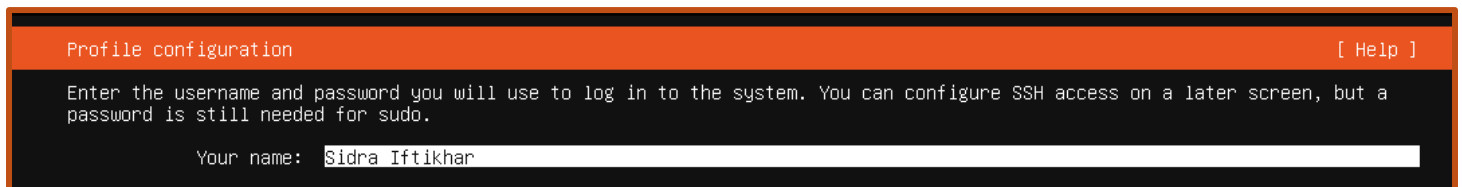
ubuntu_storage_confirmed:



Screenshot: Ubuntu storage confirmed

Task 13: Configure the Ubuntu Profile

ubuntu_actual_name:



Screenshot: Ubuntu actual name

ubuntu_server_name



Screenshot: Ubuntu server name

ubuntu_username



Screenshot: Ubuntu username

ubuntu_profile_setup

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Sidra Iftikhar

Your servers name: ubuntu
The name it uses when it talks to other computers.

Pick a username: engineer-sidra-iftikhar

Choose a password: *****

Confirm your password: *****_

Screenshot: Ubuntu profile setup

Task 14: Complete the Software Installation

ubuntu_installation_progress:

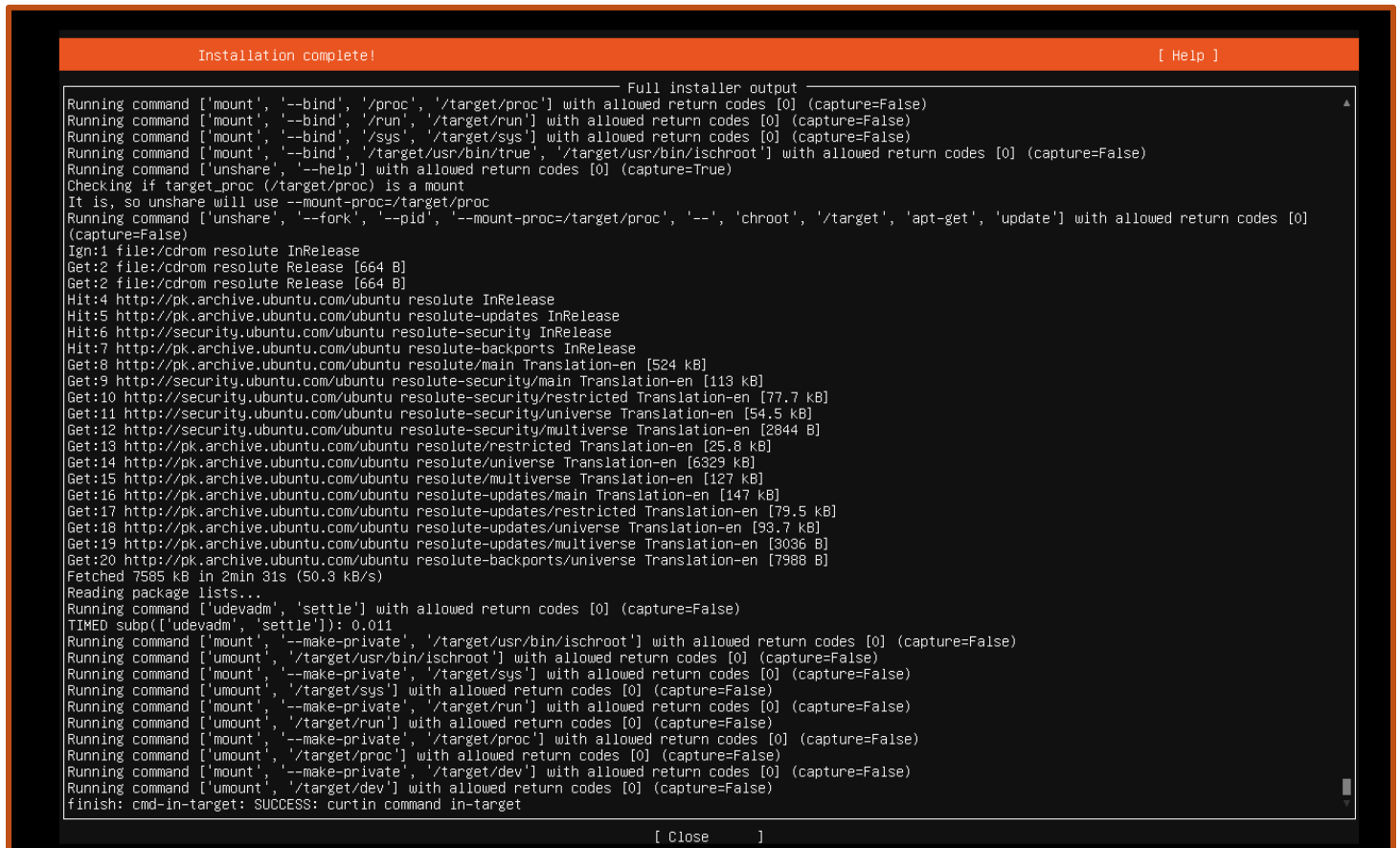
Installing system [Help]

```
subiquity/load_cloud_config/extract_autoinstall:
subiquity/Early/apply_autoinstall_config:
subiquity/Reporting/apply_autoinstall_config:
subiquity/Error/apply_autoinstall_config:
subiquity/Userdata/apply_autoinstall_config:
subiquity/Package/apply_autoinstall_config:
subiquity/Debconf/apply_autoinstall_config:
subiquity/Kernel/apply_autoinstall_config:
subiquity/KernelCrashDumps/apply_autoinstall_config:
subiquity/Zdev/apply_autoinstall_config:
subiquity/Ad/apply_autoinstall_config:
subiquity/Late/apply_autoinstall_config:
configuring apt
curtin command in-target /
```

Screenshot: Ubuntu installation progress

Task 15: Reboot and Verify the Ubuntu Identity

ubuntu_installation_complete:



```
Installation complete! [ Help ]

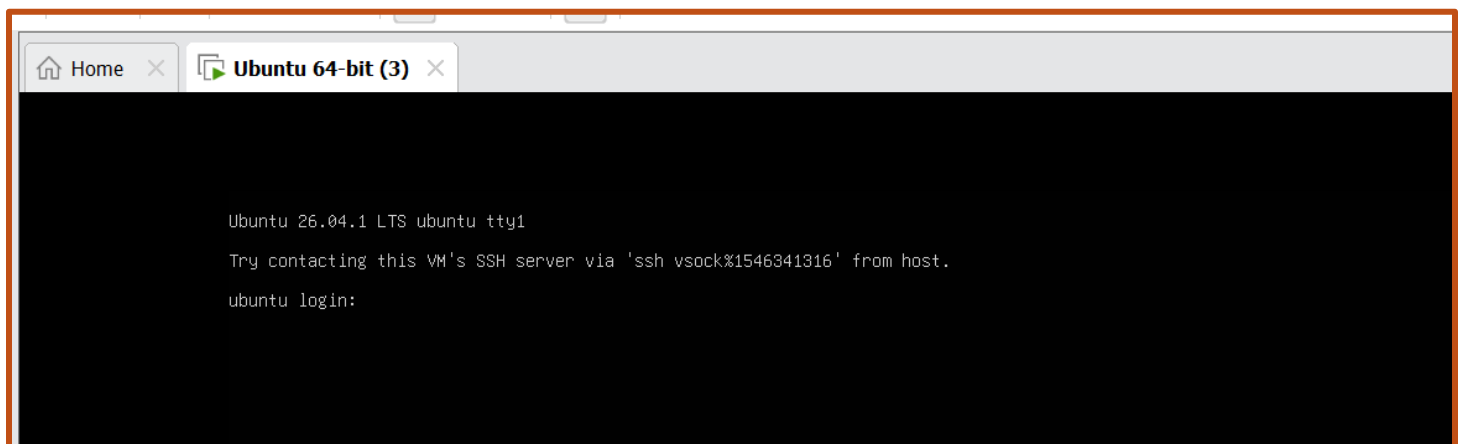
Full installer output

Running command ['mount', '--bind', '/proc', '/target/proc'] with allowed return codes [0] (capture=False)
Running command ['mount', '--bind', '/run', '/target/run'] with allowed return codes [0] (capture=False)
Running command ['mount', '--bind', '/sys', '/target/sys'] with allowed return codes [0] (capture=False)
Running command ['mount', '--bind', '/target/usr/bin/true', '/target/usr/bin/ischroot'] with allowed return codes [0] (capture=False)
Running command ['unshare', '--help'] with allowed return codes [0] (capture=True)
Checking if target_proc (/target/proc) is a mount
It is, so unshare will use --mount-proc=/target/proc
Running command ['unshare', '--fork', '--pid', '--mount-proc=/target/proc', '--', 'chroot', '/target', 'apt-get', 'update'] with allowed return codes [0] (capture=False)
Ign:1 file:/cdrom/resolute InRelease
Get:2 file:/cdrom/resolute Release [664 B]
Get:2 file:/cdrom/resolute Release [664 B]
Hit:4 http://pk.archive.ubuntu.com/ubuntu/resolute InRelease
Hit:5 http://pk.archive.ubuntu.com/ubuntu/resolute-updates InRelease
Hit:6 http://security.ubuntu.com/ubuntu/resolute-security InRelease
Hit:7 http://pk.archive.ubuntu.com/ubuntu/resolute-backports InRelease
Get:8 http://pk.archive.ubuntu.com/ubuntu/resolute/main Translation-en [524 kB]
Get:9 http://security.ubuntu.com/ubuntu/resolute-security/main Translation-en [113 kB]
Get:10 http://security.ubuntu.com/ubuntu/resolute-security/restricted Translation-en [77.7 kB]
Get:11 http://security.ubuntu.com/ubuntu/resolute-security/universe Translation-en [54.5 kB]
Get:12 http://security.ubuntu.com/ubuntu/resolute-security/multiverse Translation-en [2844 B]
Get:13 http://pk.archive.ubuntu.com/ubuntu/resolute/restricted Translation-en [25.8 kB]
Get:14 http://pk.archive.ubuntu.com/ubuntu/resolute/universe Translation-en [6329 kB]
Get:15 http://pk.archive.ubuntu.com/ubuntu/resolute/multiverse Translation-en [127 kB]
Get:16 http://pk.archive.ubuntu.com/ubuntu/resolute-updates/main Translation-en [147 kB]
Get:17 http://pk.archive.ubuntu.com/ubuntu/resolute-updates/restricted Translation-en [79.5 kB]
Get:18 http://pk.archive.ubuntu.com/ubuntu/resolute-updates/universe Translation-en [93.7 kB]
Get:19 http://pk.archive.ubuntu.com/ubuntu/resolute-updates/multiverse Translation-en [3036 B]
Get:20 http://pk.archive.ubuntu.com/ubuntu/resolute-backports/universe Translation-en [7988 B]
Fetched 7585 kB in 2min 31s (50.3 kB/s)
Reading package lists...
Running command ['udevadm', 'settle'] with allowed return codes [0] (capture=False)
TIMED subp(['udevadm', 'settle']): 0.011
Running command ['mount', '--make-private', '/target/usr/bin/ischroot'] with allowed return codes [0] (capture=False)
Running command ['mount', '/target/usr/bin/ischroot'] with allowed return codes [0] (capture=False)
Running command ['mount', '--make-private', '/target/sys'] with allowed return codes [0] (capture=False)
Running command ['mount', '/target/sys'] with allowed return codes [0] (capture=False)
Running command ['mount', '--make-private', '/target/run'] with allowed return codes [0] (capture=False)
Running command ['mount', '/target/run'] with allowed return codes [0] (capture=False)
Running command ['mount', '--make-private', '/target/proc'] with allowed return codes [0] (capture=False)
Running command ['mount', '/target/proc'] with allowed return codes [0] (capture=False)
Running command ['mount', '--make-private', '/target/dev'] with allowed return codes [0] (capture=False)
Running command ['mount', '/target/dev'] with allowed return codes [0] (capture=False)
finish: cmd-in-target: SUCCESS: curtin command in-target

[ Close ]
```

Screenshot: Ubuntu installation complete

ubuntu_login_screen:



```
Home x Ubuntu 64-bit (3) x

Ubuntu 26.04.1 LTS ubuntu tty1

Try contacting this VM's SSH server via 'ssh vsock%1546341316' from host.

ubuntu login:
```

Screenshot: Ubuntu login screen

ubuntu_terminal_login:

```
Ubuntu 26.04.1 LTS ubuntu tty1
```

```
Try contacting this VM's SSH server via 'ssh vsock%1546341316' from host.
```

```
ubuntu login: engineer-sidra-iftikhar  
Password:
```

Screenshot: Ubuntu terminal login

ubuntu_identity_verified:

```
engineer-sidra-iftikhar@ubuntu:~$
```

Screenshot: Ubuntu identity verified

Task 16: Find the Ubuntu Server IP Address

ubuntu_ip_addr_command:

```
engineer-sidra-iftikhar@ubuntu:~$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000  
    link/ether 00:0c:29:2b:4b:c4 brd ff:ff:ff:ff:ff:ff  
    altname enp2s1  
    altname enx000c292b4bc4  
    inet 192.168.124.132/24 metric 100 brd 192.168.124.255 scope global dynamic ens33  
        valid_lft 1048sec preferred_lft 1048sec  
    inet6 fe80::20c:29ff:fe2b:4bc4/64 scope link proto kernel_ll  
        valid_lft forever preferred_lft forever  
engineer-sidra-iftikhar@ubuntu:~$
```

Screenshot: Ubuntu ip addr command

ubuntu_ip_address

```
inet 192.168.124.132/24 metric 100 brd 192.168.124.255 scope global dynamic ens33  
    valid_lft 1048sec preferred_lft 1048sec  
inet6 fe80::20c:29ff:fe2b:4bc4/64 scope link proto kernel_ll  
    valid_lft forever preferred_lft forever  
engineer-sidra-iftikhar@ubuntu:~$
```

Screenshot: ubuntu_ip_address

Task 17: Connect to Ubuntu Server from Windows

windows_ssh_command:

```
Warning: Permanently added '192.168.124.132' (ED25519) to the list of known hosts.  
engineer-sidra-iftikhar@192.168.124.132's password:  
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)
```

Screenshot: Windows ssh command

windows_ssh_identity

```
Warning: Permanently added '192.168.124.132' (ED25519) to the list of known hosts.  
engineer-sidra-iftikhar@192.168.124.132's password:  
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)
```

Screenshot: Windows ssh identity

Task 18: Prepare Lab1 Solution PDF

solution_title_page



Lab 01: GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Name: Sidra Iftikhar

Reg no: FA24B1-SE-063

Course code: SE-665

Section: A

GitHub Username: engineer-sidra-iftikhar

Screenshot: Solution title page